

Effects of Foreground Mitigation Placement in the HERA Analysis Pipeline on 21 cm Signal Recovery.

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Research Area

This project lies in the field of radio astronomy data analysis, specifically in 21 cm cosmology and Epoch of Reionisation (EoR) signal detection using the Hydrogen Epoch of Reionisation Array (HERA (Hydrogen Epoch of Reionization Array) framework). It focuses on signal extraction in the presence of strong astrophysical foreground contamination, a central challenge in low-frequency cosmological observations. The work connects statistical signal processing, and computational astrophysics.

Introduction

The Hydrogen Epoch of Reionisation Array (HERA (Hydrogen Epoch of Reionization Array)) aims to detect the highly redshifted 21 cm hydrogen line from the early universe, probing cosmic structure formation and the Epoch of Reionisation. However, the cosmological signal is buried beneath astrophysical foregrounds that are several orders of magnitude stronger.

A key challenge is separating the faint cosmological signal from bright foreground emission without introducing bias or excessive signal loss, where the placement of foreground mitigation within the analysis pipeline can significantly affect the fidelity of the recovered information.

Aim of the Project

This project aims to investigate how the placement of foreground mitigation within the HERA data analysis pipeline influences the recovery of the 21 cm cosmological signal, and to quantify the resulting bias and information loss using both traditional filtering approaches and data-driven methods.

Project Activities

This project may include the following activities:

- Literature review of 21 cm foreground mitigation (e.g. delay filtering, Gaussian processes).
- Analytical modelling of signal–foreground separation in simplified pipelines
- Numerical simulations of mock HERA visibility data
- Application of foreground removal at different pipeline stages
- Data-driven mitigation approaches
- Quantification of signal recovery (power spectrum bias, residuals)
- Comparison of pipeline configurations for information loss and fidelity

References

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- [3] HERA Collaboration technical overview papers.